PM Graded Quiz M3

Question 1

In using the Black-Litterman approach as a reverse optimization technique, what formula represents the error in the implied expected return, according to the required reading?

- $(\Sigma_1 - \Sigma_0)w_0$.
- $(\Sigma_1 - \Sigma_0)w_M$.
- $w_M^T(\Sigma_1 - \Sigma_0)w_0$.
- ◉ $(\Sigma w_0 - \Sigma_0)w_0$.

Question 2

Assume that we have 4 assets (A, B, C, D). The investor's views are:

- Asset B will achieve a return of 5% more than asset C. The manager has confidence his can be expressed by $\omega_1 = 4\%$.
- Asset D will achieve a return of 10% more than in the previous period. His confidence is $\omega_2 = 6\%$.

- $P = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$.
  $\Omega = \begin{pmatrix} 0.04 & 0 \\ 0 & 0.06 \end{pmatrix}$.
- $P = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$.
  $\Omega = \begin{pmatrix} 0.04^2 & 0.024 \\ 0.024 & 0.06^2 \end{pmatrix}$.
- $P = \begin{pmatrix} 0 & 1/2 & -1/2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$.
  $\Omega = \begin{pmatrix} 0.04^2 & 0 \\ 0 & 0.06^2 \end{pmatrix}$.
- ◉ $P = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$.
  $\Omega = \begin{pmatrix} 0.04^2 & 0 \\ 0 & 0.06^2 \end{pmatrix}$.

Question 3

The picture below represents a simulation in the space $(0,1) \times (0,1)$. Do you think that this picture represents a quasi random simulation, a Monte Carlo simulation, or both?


Source: Wikipedia

- It is neither. This is a simulation from a Markov Chain Monte Carlo.
- It is very likely a quasi-random simulation. The probability that a Monte Carlo simulation generates such a regular grid is extremely low.

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- It is a quasi-random simulation. Monte Carlo simulations cannot generate such regular outcomes.
- (•) It is a Monte Carlo simulation. A quasi-random simulation would be more diffused inside the square.

Question 4

A quantitative analyst just created the following table for the portfolio manager in charge of Mrs. Smith's accounts:

				Asset Weight				
Corner P.folio	Expected Return	Expected Std. Dev	Sharpe Ratio	Asset 1	Asset 2	Asset 3	Asset 4	Asset 5
1	17.4%	20.5%	0.85	0.00	0.00	1.00	0.00	0.00
2	16.2%	16.0%	1.01	0.00	0.00	0.85	0.00	0.15
3	13.6%	13.9%	0.98	0.22	0.00	0.55	0.00	0.23
4	12.6%	13.5%	0.93	0.21	0.00	0.50	0.05	0.24
5	10.6%	11.5%	0.92	0.10	0.49	0.21	0.00	0.20
6	9.5%	10.9%	0.87	0.00	0.87	0.04	0.00	0.09

Mrs. Smith is requiring a real return of 14%. Inflation is currently estimated at 1%. How should the portfolio manager invest among the different corner portfolios? (Note: Expected returns in the table are all nominal.)

What is the weight of asset 1 and asset 5 in the portfolio built for Mrs. Smith?

- Asset 1: 0.118 Asset 5: 0.193
- Asset 1: 0.186 Asset 5: 0.215
- Asset 1: 0.102 Asset 5: 0.187
- (•) Asset 1: 0.205 Asset 5: 0.188

Question 5

In the JLZ model, what are the priors for

$f|h$ and h?

- $f|h \sim N(m_f, V_f/h)$ and $h \sim \text{Gamma}(p, q)$.
- $f|h \sim N(m_f, V_f/h)$ and $h \sim N(p, q)$.
- $f|h \sim \text{Gamma}(m_f, V_f/h)$ and $h \sim N(p, q)$.
- (•) $f|h \sim N(m_f, V_f/h)$ and $h \sim \text{Gamma}(2, q)$.

Question 6

When using the Black-Litterman model, what are the μ and Ψ to use in the optimization program:

$$\begin{aligned} &\max_w w^T\mu - \frac{1}{2}w^T\Psi w \\ &\text{subject to } w^t\iota = 1. \end{aligned}$$

- $\mu = [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\pi + P^T\Omega^{-1}q]$;

  and

  $\Psi = \Sigma + [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}$.

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- $\mu = [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\pi + P^T\Omega^{-1}q]$;

  and

  $$\Psi = [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}.$$

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- $\mu = [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\pi + P^T\Omega^{-1}q]$;

  and

  $$\Psi = \Sigma + [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P].$$

Question 7

Why in the lesson notes was it claimed that for pension funds or life insurance companies the traditional mean-variance optimization is **not** always enough as a way to manage their financial portfolios?

- These companies are often mandated by laws to be more conservative in the way they invest their funds, and mean-variance optimization cannot guarantee a low risk.
- These companies cannot assume the same financial risks as regular investors or mutual funds.
- These companies can only invest in bonds and not equities so they use duration as a measure of risk and not volatility.
- These companies have spending needs, which require certain amounts of liquidity over different periods and mean-variance optimization cannot always account for that.

Question 8

Consider the following table of assets and their expected returns:

Asset	A	B	C	D
Exp. Return	10%	11%	9%	15%

However, the constraints for the weights are now given by

$$0 \leq w_A \leq 0.5, 0 \leq w_B \leq 0.2, 0 \leq w_C \leq 0.3, 0 \leq w_D \leq 0.4$$

together with the requirement that

$$w_A + w_B + w_C + w_D = 1$$

What is the value assigned to asset B in the second step of the critical line algorithm?

And what is the value assigned to asset A, also in the second step of the same algorithm?

- $w_B = 0.2\, w_A = 0.5$
- $w_B = 0.1\, w_A = 0.5$
- $w_B = 0.0\, w_A = 0.5$
- $w_B = 0.2\, w_A = 0.4$

Question 9

Assume that the factor exposure of the portfolio is expressed via the matrix

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$$B^T = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Furthermore, we have estimated the variance-covariance matrix as

$$\hat{\Sigma} = \begin{pmatrix} 0.05 & 0.02 & 0 & 0 \\ 0.02 & 0.06 & 0 & 0 \\ 0 & 0 & 0.07 & 0 \\ 0 & 0 & 0 & 0.09 \end{pmatrix}$$

and our factor view vector is

$$\lambda = \begin{pmatrix} 0.02 \\ 0.03 \end{pmatrix}.$$

What is the Black-Litterman alpha?

- $\begin{pmatrix} 0.00103 \\ 0.00197 \\ 0.00000 \\ 0.00103 \end{pmatrix}$.
- $\begin{pmatrix} -110.207 \\ 179.364 \\ 0.000 \\ 9.25926 \end{pmatrix}$.
- $\begin{pmatrix} 0.475 \\ 0.075 \\ 0.105 \\ 0.075 \end{pmatrix}$.
- ◉ $\begin{pmatrix} 0.075 \\ -0.475 \\ 0.000 \\ 0.075 \end{pmatrix}$.

Question 10

Which of the following is a benefit of using the Black-Litterman model? Choose all that apply:

- ☑ The Black-Litterman model produces overall better portfolios in terms of performance.
- ☑ Optimal portfolio is less sensitive to changes in expected return estimates.
- ☑ Optimal portfolio is less likely to produce very high and very negative weights for some of the assets (if we do not add any constraint on size of positions).
- ☑ Optimal portfolio tends to exhibit less concentrated positions.

Question 11

Match the step number to its correct action in implementing PSO as described in the lesson notes.

A) Measure the required probability and maximize the objective function

B) Select the investment universe and indicate allocation constraints

C) Specify the objective function in terms of probability of a specific return and a risk measure

D) Reduce the admissible asset allocations to the set of risk-adequate portfolios with respect to the investor's risk-profile. In other words, we filter out those portfolio that do not comply with our ambitions.

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E) Set up the scenarios to simulate

Step 4	A
Step 2	B
Step 8	C
Step 5	D

Question 12

Suppose we are given the following assets with their respective expected returns:

Asset	1	2	3	4
Exp. Return	10%	8%	11%	13%

and that the constraints for the problem are:

$$0 \leq w_1 \leq 0.5, 0 \leq w_2 \leq 0.4, 0 \leq w_3 \leq 0.25, 0 \leq w_4 \leq 0.3.$$

and

$$w_1 + w_2 + w_3 + w_4 = 1.$$

If we perform the first two steps of the critical line algorithm, identify the classes F and B.

- F={1,2,3} B={4}
- F={2,3,4} B={1}
- F={1} B={2,3,4}
- F={1,2} B={3,4}

Question 13

What is the purpose of using a gamma distribution when the paper uses:

$$f|h \sim N(m_f, V_f/h) \quad \text{and} \quad h \sim \text{Gamma}(p, q)$$

- The Gamma-Normal prior is a conjugate prior, and the posterior distribution will have a t distribution. The choice of the prior will not affect the results we find.
- The Gamma-Normal prior is a conjugate prior, and the posterior distribution will have the same type of distribution as the prior. The results do not depend on the choice of the prior.
- The Gamma-Normal prior is a conjugate prior and the posterior distribution will be a gamma distribution. Unfortunately, the result of the analysis depends on the prior used.
- The Gamma-Normal prior is a conjugate prior, and the posterior distribution will have the same type of distribution as the prior. Unfortunately, the results change if we change the prior.

Question 14

Which of the following statements is **false**?

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- The market portfolio can only be a world market portfolio like MSCI or GSMI.
- We can use a local market portfolio as long as we remember to determine its Sharpe ratio and its portfolio weights.
- To use the Black-Litterman model, the user must define a market portfolio and needs to compute the weights of the assets with respect to that market portfolio.
- If an individual portfolio manager has no views, they will hold the market portfolio.

Question 15

In dealing with an optimization problem like:

$$\max_{x_1,\dots,x_N} \{\min f(x_i), \sum_{i=1}^{N} r_i x_i\}$$
$$\text{s.t. } (x_1, \dots, x_N) \in \mathcal{X}.$$

How can the problem be described?

- A maxmin optimization problem
- A bi-criteria optimization problem
- A minmax optimization problem
- A regular optimization problem

Question 16

Does the critical line algorithm work for the Black-Litterman model?

- Yes, but only if there are no inequality constraints; otherwise, the Black-Litterman model requires a different approach.
- No, the model was developed by Markowitz for the classical mean-variance optimization problem.
- Yes, using the Black-Litterman model only changes the estimates of μ and Σ, and the rest stays the same.
- No. They are very different models that require different parameters to be implemented.

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